

Test Plan

Airline Management System

Version 1.0

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1. Test Plan Identifier:

Airline Management System -Version 1.0

2. References:

IEEE-829 test plan format is used for this test plan document, and SRS of the Airline Management System (AMS) of our group is used as the basis for this test plan.

3. Introduction:

The main purpose of the test plan for the Airline Management System is to discuss the testing details of the use cases and test cases of the Airline Management System.

It is the Master Plan for the Airline Management System. This plan will address only the items and elements that are related to the AMS , both directly and indirectly affected items will be addressed. This plan's primary focus is to testify that the information and its functionalities are working as required in the project statement and the level of availability of details.

Scope of testing: The project will have 2 testing levels UnitTesting(manual and automatic). The details for each level are addressed in the approach section and will be further defined in the level of specific plans.

4. Test Items:

Each Django app consists of Urls, views and models. We check all those for every app. The following is a list of those items, that are tested for every app :

- **Airlines**

- Models:

- Airport
 - Airline
 - AirplaneType
 - SeatType
 - Flight
 - Schedule
 - Trips

- **AMS**

- Urls:

- Homepage
 - Help
 - About

- Views:

- homepage(request)
 - help_main(request)
 - about(request)

- **UserAccounts**

- Models:

- User
 - Passenger
 - Manager

- Urls:

- Choose
 - Passenger_signup
 - Manager_signup
 - Login
 - Logout
 - Logout_confirm

- Views:

- Passenger_signup(CreateView)
 - Manager_signup(CreateView)
 - Choose_view(request)
 - Login_view(request)
 - Logout_confirm(request)
 - Logout_view(request)

- **Passengers**

- Models:

- Tickets

- Urls:

- Passenger
 - Book
 - Balance
 - Details
 - Trips
 - Search
 - Flight
 - Ticket
 - Modify_ticket
 - Del_ticket
 - Detail
 - choose_seat

- Views:

- Passenger_homepage(request)
 - Ticket_detail(request, slug)
 - Add_detail(request)
 - Add_balance(request)
 - Book_ticket(request)
 - Search_flight(request)
 - See_tickets(request)
 - Flight_detail(request, slug)
 - Delete_ticket(request, slug)
 - Modify_ticket(request, slug)
 - Details_of_flight(request, slug)
 - Choose_seat(request, slug)

- **Managers**

- Urls:

- Manager
 - Check
 - Change
 - Add
 - Delete
 - Profit
 - Search
 - View_occupancy
 - Costs
 - Modify
 - Del

- Views:
 - Manager_homepage(request)
 - Check_occupancy(request)
 - View_occupancy(request)
 - Change_schedule(request)
 - Get_profits(request)
 - Change_costs(request)
 - Modify_costs(request)
 - Delete_flights(request)
 - Search_flights(request)
 - Add_flight(request)
 - Del_flight(request, slug)

5. Software Risk Issues:

AMS has some Risk issues:

- Manager has power to delete and add Flights and he should be aware of Algorithm used to check if new Schedule is Possible or not
- If any models are changed in future, the database need to be deleted and it can be recreated by running a simple script

6. FeaturesTo be Tested:

The following are the list of features which are to be focussed

- Features available to Passenger:
 - Signup
 - Login
 - Logout
 - Search Flights
 - Book Ticket
 - Choose Seat Number
 - View and Add Balance
 - View Upcoming Trips
 - Modify or Cancel Ticket
- Features available to Manager:
 - Signup
 - Login
 - Logout

- Check Occupancy Rate
- Change Schedule
- Add Flight
- Delete Flight
- Get Profits
- Change Airplane Costs

7. Features Not to be tested:

The following is a list of the areas that will not be specifically addressed. All testing in these areas will either be indirect as a result of other testing efforts or are implemented by the web framework Django used or are not fully implemented

- Validation of Credit Card Details
- Validation of User Details as phone number or email id
- Size allocation and data types of Class Attributes

8. Approach :

a. Unit Test Cases

These are performed on each app of Django Project

- The urls are checked for valid function calls when their interface is loaded
- All the model fields are tested for their labels, choices(for enumeration types), type of relation with other models(one to one, many to one) and object representation for their golden outputs.
- All the view functions used to render the HTML pages and receiving responses in the form of requests are checked for proper redirection to valid HTTP responses along with the required data(context).

The team will meet once every day to evaluate progress to date and to identify error trends and problems as early as possible. These meetings will be scheduled at the same time every day. Additional sessions can be called as required for emergencies in order to make some changes in the project.

9. Item Pass/Fail Criteria:

If Test Cases Fail in any of the following areas, we declare that the system has failed:

- Signup System
- Login System
- Ticket Booking and Cancellation System
- Searching Flights System
- Checking Profits and Occupancy System
- Changing Schedule System
- Managing Costs System

If none of these Fail, we declare that AMS has passed

10. Suspension Criteria and Resumption Requirements

a. Suspension Criteria:

This Criteria originates from ticket booking or search flights, if error occurs in these test cases, we temporarily stop the testing process and start debugging process as Passengers are the primary users of the System and these functionalities are required

b. Resumption Criteria:

When a new version of the system is transmitted to the test group after a suspension of testing has occurred, all previous tests will be rerun to ensure program changes have not inadvertently affected other portions of the program.

11. Test Deliverables:

- Unit test plan
- Test cases
- Test Outputs
- Error logs and execution logs

12.Environmental Needs:

For Testing of AMS, the following requirements must be met by the environment:

- Django 2.2 or above
- Python 3 or above
- Site Packages : django_extensions

13.Staffing and Training Needs:

It is preferred that there will be at least one full time tester assigned to the project for system/integration and acceptance testing phases of the project. The following areas are needed to be addressed in terms of training:

- The tester should be trained how to interact with the frontend interface of AMS
- They should be trained to run a script to recreate DataBase
- The Tester should be trained on basic operations of Django Admin
- They should be also be trained logistics of Ticket Booking and expected input-outputs of AMS so these criteria can be tested

14. Responsibilities:

Responsibilities	Team	Client
SRS Documentation and Implementation	X	
Test Plan and Test Case	X	
Unit Test documentation and execution	X	
System Design Reviews	X	X

The team will be responsible for verifying and accepting all unit test plans and documentation and implementation. The entire project team will participate in reviewing the system and detail designs and reviewing any change requests generated by the user or defects discovered during development and testing. All these works will be equally divided among the group

15. Schedule:

Time has been allocated viewing the project plan for the following reviewing activities. The specific dates and times for each activity are defined in the project plan timeline.

- Review and documentation of the Software Requirements Specifications (SRS) by the team and create the inventory classes, methods, and objectives
- Development of the Master Plan and documentation of the test cases and class diagrams.
- Review of the implementation and the unit test and test cases.
- Presentation of the final output of the project.

16. Planning Risks and Contingencies:

- Delay in the training of application or tools :
We are making a web-based application that is very new to the team, and using django framework as middleware is also new to get trained for that software and tools takes time, which may cause a delay in submission
- Change in original requirements or design : This test plan and SRS is being made without code and may have some differentiation with the implementation Some of the mentioned conditions may change at the time of execution and testing according to the team's convenience and resources availability.

However, the team's time availability may balance the problem mentioned above and submit the test plan and implementation and unit test on time.

17. Approvals:

The Test Plan is Submitted for review and approval by the professor and mentor.

Project Allocated By:

- **Professor :** Sourangshu and Abir Das
- **Developers/Programmers/Testers :** Jaswanth,Sushanth,Sathvik
- **Mentor :** Mr.Shiva Kumar Veldi

18. Glossary:

Title	Description
AMS	Airline Management System
App	A small library representing a discrete part of a larger Django project
Django	Django is a Python-based free and open-source web framework that follows the model-template-views architectural pattern.
GUI	Graphical User Interface
Manager	Secondary User of the System
Passenger	Primary User of the System
SRS	Software Requirements Specification